

CLIMINVEST: Extraction Agent and replication



Climinvest



Ministero delle Imprese
e del Made in Italy



Funded by
the European Union
NextGenerationEU

Climinvest - Extraction Agent (starting data)

Projects on climate change adaptation

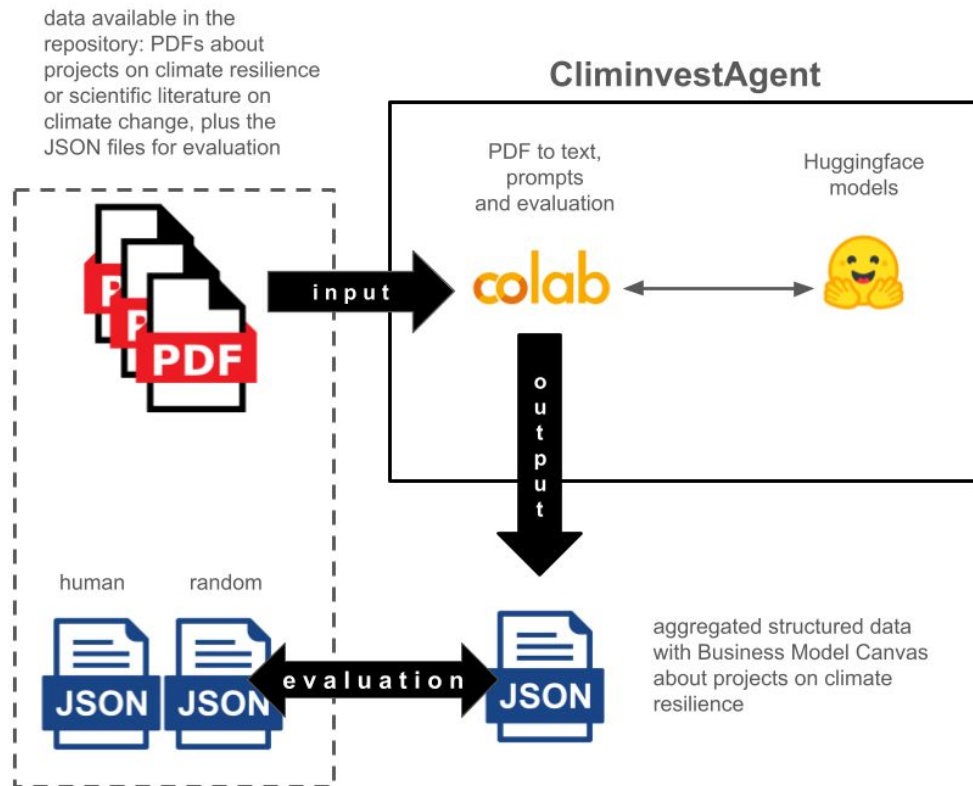
Reports

Tools as outcomes of projects



unstructured → structured

Climinvest - Extraction Agent (dev & test)

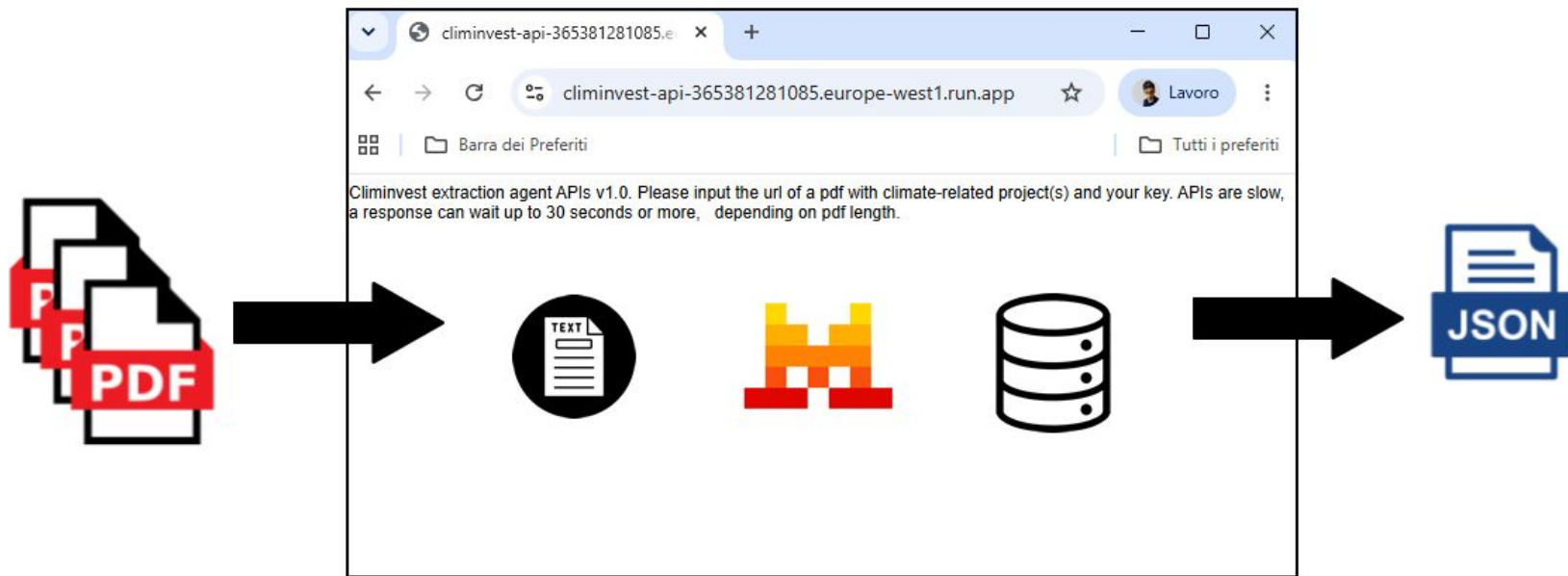


Climinvest - Extraction Agent evaluation

Results

Model	Prompt	Trial	T	CS.n	CS.h	Cover	Spar	Δ CS	Prec	Rec	F1
Claude 3.5 Sonnett	P1	1	36	0.790	0.799	0.095	0.125	0.009	0.0000	0.0000	0.0000
		2	37	0.764	0.783	0.095	0.175	0.019	0.0678	0.0317	0.0432
		3	36	0.774	0.785	0.143	0.083	0.011	0.0143	0.0079	0.0102
	P2	1	55	0.732	0.747	0.238	0.190	0.015	0.0000	0.0000	0.0000
		2	53	0.729	0.746	0.238	0.130	0.017	0.0084	0.0159	0.0110
		3	48	0.761	0.734	0.190	0.113	-0.027	0.0000	0.0000	0.0000
GPT oss	P1	1	23	0.823	0.760	0.429	0.805	-0.063	0.0000	0.0000	0.0000
		2	21	0.864	0.815	0.333	0.333	-0.049	0.0000	0.0000	0.0000
		3	21	0.840	0.804	0.381	0.778	-0.036	0.0000	0.0000	0.0000
	P2	1	21	0.836	0.817	0.286	0.300	-0.019	0.0000	0.0000	0.0000
		2	22	0.820	0.804	0.190	0.263	-0.016	0.0000	0.0000	0.0000
		3	18	0.765	0.776	0.190	0.300	0.011	0.0000	0.0000	0.0000
Mistral 3 medium	P1	1	52	0.830	0.807	0.143	0.417	-0.023	0.0101	0.0079	0.0089
		2	52	0.830	0.807	0.095	0.375	-0.023	0.0103	0.0079	0.0090
		3	42	0.830	0.807	0.143	0.400	-0.023	0.0123	0.0079	0.0097
	P2	1	51	0.726	0.746	0.333	0.221	0.020	0.0168	0.0317	0.0220
		2	49	0.737	0.762	0.286	0.217	0.025	0.0140	0.0238	0.0176
		3	45	0.723	0.744	0.286	0.217	0.021	0.0127	0.0238	0.0165

Climinvest - Extraction Agent (APIs)



Climinvest - Extraction Agent (json schema part 1)

```
{  
  "project": "Project Reef", "country": "Philippines", "year": "2026",  
  "target_events": ["typhoons", "storm surges", "power outages"],  
  "project_summary": "Project Reef focuses on developing solar-powered microgrids with battery storage to provide reliable energy to  
vulnerable coastal communities in the Philippines.",  
  "main_project_result": "Provides 24/7 power to 40,000 residents, enhancing resilience to typhoons and storm surges.",  
  "discount_rate_cost_effectiveness": null,  
  "keywords": ["solar microgrids", "typhoon resilience", "coastal communities", "Philippines", "renewable energy",  
  "energy security" ],  
  "KPIs_to_monitor": [ "energy generation", "number of households with reliable power", "reduction in power outages", "community  
feedback" ],  
  ...  
}
```

Climinvest - Extraction Agent (json schema part 2)

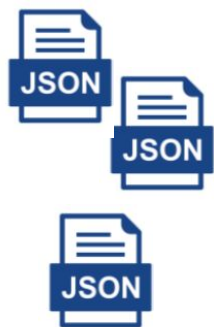
```
...  
"bm_key_partners": [ "Local government", "Energy technology providers", "Community organizations"],  
"bm_key_resources": [ "Solar technology", "Battery storage systems", "Local workforce", "Government incentives" ],  
"bm_key_activities": [ "Installation of solar microgrids", "Community training on energy use", "Maintenance of energy systems" ],  
"bm_customer_segments": [ "Coastal communities", "Local businesses" ],  
"bm_value_propositions": [ "Reliable energy supply", "Reduced dependency on diesel generators", "Enhanced community  
resilience"],  
"bm_customer_relationships": [ "Community energy committees", "Regular maintenance checks", "Feedback sessions" ],  
"bm_channels": [ "Local government collaborations", "Community centers", "Energy cooperatives"],  
"bm_cost_structure": [ "Technology procurement", "Installation costs", "Maintenance expenses", "Community training programs"],  
"bm_revenue_streams": [ "Government subsidies", "International climate funds", "Community energy fees"],  
...  
...
```

Climinvest - Extraction Agent (json schema part 3)

```
...  
"sustainability_strategy": "Ensuring long-term operation and maintenance of microgrids through local training and partnerships with energy providers.",  
"actions_to_reach_bankability": "Demonstrating the reliability and cost savings of solar microgrids to attract further investment and government support.", "legal_references": [ "Philippine energy regulations", "International renewable energy standards"],  
"regulation_framework": "Adherence to local energy regulations and international renewable energy guidelines to ensure project feasibility and community benefits."  
}
```

Climinvest Extraction Agent: strategy for replication

Projects from the
extraction agent
(PROJECTS DB)



Local regulations
and policies
(LEGAL DB)

Big document
with projects,
local regulation
and policies



attach

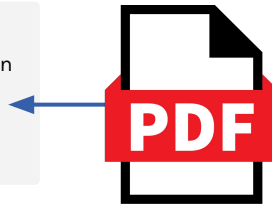


get

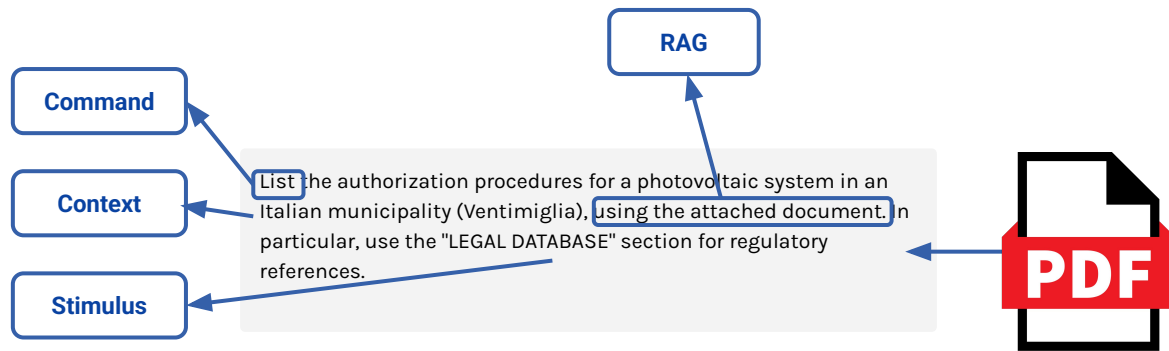
Suggestions
for locally
replicable
climate-related
projects

CLIMINVEST - Mindy - RAG

List the authorization procedures for a photovoltaic system in an Italian municipality (Ventimiglia), using the attached document. In particular, use the "LEGAL DATABASE" section for regulatory references.



CLIMINVEST - Mindy - RAG



Authorization procedures for photovoltaic systems in the Municipality of Ventimiglia

Brief overview: For the installation of photovoltaic systems in the Municipality of Ventimiglia, authorization procedures depend on the type and location of the system, as well as the presence of landscape constraints. The relevant regulatory framework includes European and national laws, supplemented by regional and municipal regulations—specifically Ventimiglia's Municipal Urban Plan (PUC).

 Key regulatory framework: Authorization procedures for photovoltaic systems in Italy are governed by a complex regulatory framework that includes:

List the authorization procedures for a photovoltaic system in an Italian municipality (Ventimiglia), using the attached document. In particular, use the "LEGAL DATABASE" section for regulatory references.

- Legislative Decree 190/2024 (Consolidated Act on Renewables): This regulation, which entered into force on December 30, 2024, and was amended in December 2025, aims to simplify and streamline the regulatory framework for renewable energy systems. It establishes three main administrative regimes:
 - Free activity: for minor or specific interventions that do not require authorization.
 - Simplified Authorization Procedure (PAS): for medium-sized systems or interventions that do not require a Single Authorization.
 - Single Authorization (AU): for more complex projects, issued by the Region or delegated provinces. Renewable energy systems are defined as works of public utility that are urgent and cannot be deferred. Regions and local authorities were granted 180 days from the entry into force of the decree to align their regulations.
- Ministerial Decree of June 21, 2024 (Suitable Areas Decree): Entered into force on July 3, 2024, this decree establishes national criteria for identifying areas suitable and unsuitable for the installation of renewable energy systems, providing the basis for accelerating the authorization process. Regional resolutions (2024–2026) implement the actual mapping of these areas.
- Presidential Decree 380/2001 (Consolidated Law on Building Regulations): Governs building activities in Italy, including building permits and administrative procedures. It has been amended over time to streamline procedures and incorporate sustainability and energy efficiency requirements. It determines whether a photovoltaic installation falls under "free building activity" (exempt from permits), the Simplified Authorization Procedure (PAS), or requires a full building permit.
- Legislative Decree 42/2004 (Code of Cultural Heritage and Landscape): The regulatory framework for the protection and management of Italy's cultural and landscape heritage. It establishes authorization requirements for interventions that may affect protected areas or assets. Strict authorization procedures apply to photovoltaic systems, particularly those installed on buildings or in areas subject to landscape restrictions.

CLIMINVEST - Mindy - RAG

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⚠ Critical distinctions and the impact of Ventimiglia's PUC: Ventimiglia's Municipal Urban Plan (PUC)—approved via City Council Resolution No. 101 on December 15, 2008—is the urban planning instrument governing all construction and land-use activities within the municipality. The PUC classifies the territory into zones, each with specific regulations regarding permitted interventions and restrictions.

- Landscape restrictions: The PUC incorporates landscape restrictions established by Legislative Decree 42/2004 (Articles 136 and 142), which affect extensive areas of the Ligurian coast and hinterland, including parts of the Municipality of Ventimiglia. Installing photovoltaic systems in these areas requires landscape authorization from the Superintendency.
- In areas classified as "suitable" under Article 20 of Legislative Decree 199/2021, the opinion of the Ministry of Culture—while still mandatory—is no longer binding (as clarified by Circular No. 2 of January 29, 2026). However, the Superintendency must still assess the project's impact.
- For installations on buildings or in areas not classified as suitable, landscape authorization remains fully binding.
- PUC Zoning: The PUC's Technical Implementation Rules (NTA) define various zoning categories relevant to photovoltaic systems:
- Historic zones of high landscape value (Art. 20(c) NTA): Any external modification, including the installation of photovoltaic panels, is subject to strict aesthetic and architectural requirements and requires landscape authorization. "Ventimiglia Città Alta" Zone (Art. 20(e) NTA): The historic urban fabric imposes additional constraints on roof modifications.
- Areas of environmental and landscape value (Art. 19(c) NTA): Ground-mounted installations may be restricted or prohibited.
- Agricultural conservation zones (Art. 21(f) NTA): Agrivoltaic or ground-mounted installations must comply with both agricultural land-use regulations and landscape requirements.
- Zones of scattered settlement with historical and landscape value (Art. 20(g) NTA): Installations on existing buildings require careful integration with the traditional built environment.
- Hydrogeological constraints: The PUC also incorporates constraints derived from the Roia River Basin Plan, imposing hydrogeological risk limitations that may affect the siting of ground-mounted photovoltaic systems in flood-prone areas.

Thank You

Questions?